

Problem 1.6

The formula

$$M^{(k)}(0) = \int_0^1 s_n^k(\omega) d\omega \quad (1)$$

follows from the k -th derivative of $e^{\lambda t}$ being $\lambda^k e^{\lambda t}$ and taking $\lambda = s_n(\omega)$. Note that interchanging differentiation and integration is warranted, since the integrand is piecewise constant in ω . We now prove (1.16), (1.18) and (1.28):

1. (1.16): Note that by $M(t) = (\cosh t)^n$ one has $M^{(1)}(t) = n(\cosh t)^{n-1} \sinh t$. Choosing $t = 0$ and using (1) with $k = 1$ yields

$$\int_0^1 s_n(\omega) d\omega = 0$$

2. (1.18) : We apply the same chain of thought. Using the shape of $M^{(1)}$ we calculate

$$M^{(2)}(t) = n(n-1)(\cosh t)^{n-2}(\sinh t)^2 + n(\cosh t)^n$$

and again using (1) with $k = 2$ and taking $t = 0$ yields

$$\int_0^1 s_n^2(\omega) d\omega = n$$

3. (1.28) : We calculate $M^{(3)}$ using the form of $M^{(2)}$

$$M^{(3)}(t) = n(n-1)(n-2)(\cosh t)^{n-3}(\sinh t)^3 + 2n(n-1)(\cosh t)^{n-1} \sinh t + n^2(\cosh t)^{n-1} \sinh t.$$

We now calculate $M^{(4)}(0)$, which by the form of $M^{(3)}(t)$ yields

$$M^{(4)}(0) = 2n(n-1)(\cosh 0)^n + n^2(\cosh 0)^n = n + 3n(n-1).$$